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Palaeomagnetism and palaeogeography of Mongolia in the Cretaceous, Permian and Carboniferous—preliminary data

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Abstract

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The paper reports on the results of palaeomagnetic stability tests of oriented samples of lithologically different types of Cretaceous, Permian and Carboniferous rocks from northern and southern Mongolia. The Permian andesites and tuffites from three localities (Barun Buren, Tel Tin Gol, and Chulut Tsagan Del) have yielded interesting results. Groups of normally and reversely polarized andesite tuffs were found even within a single outcrop by means of thermal field demagnetization and multicomponent analysis of the magnetization. With a view to the number of samples (177) suitable for analysis and the number of localities (9), the reported palaeomagnetic data should be considered as preliminary. The interpreted results indicate a different palaeogeographic evolution of northern and southern Mongolia relative to the Siberian platform and North China in the Permian and Carboniferous. On the other hand, similar apparent polar wander paths for North China (Lin et al., 1985) and the studied regions of northern and southern Mongolia indicate similar palaeogeographic evolution in the Permian to Carboniferous.

Introduction

The palaeogeographic evolution of Mongolia during the upper Palaeozoic and Mesozoic is a very important problem, particularly with respect to its relationship with the Siberian platform to the north and the North China block to the south. Therefore samples of various Cretaceous, Permian and Carboniferous rocks were collected in northern Mongolia in the Barun Buren region, in central Mongolia, in the neighbourhood of the Chulut Tsagan Del ore deposit and in southern Mongolia in the Gobi Desert during the geophysical expedition undertaken in 1980 and 1981.

The objective of this stage of the research was: (a) to analyse the results of the palaeomagnetic investigations of the sampled rocks, (b) to designate the types of rocks suitable for further palaeomagnetic research with a view to determining the appropriate palaeopoles and palaeolatitudes in central Asia.

The palaeomagnetic and rock magnetic investigations were carried out in the Průhonice Palaeomagnetic Laboratory of the Prague Division of Geofyzika, N.C.

Geological setting and sampling

The Chulut Tsagan Del locality in central Mongolia is made of Precambrian phyllites, limestones, porphyrites, Permian andesite tuffs and sandstones containing Jurassic granite intrusions (Table 1).

The Permian volcanics, which have a lava-like appearance in the field, are mostly pyroclastics. They include primarily lithoclastic and crystallo-

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TABLE 1

Review of collected oriented samples in Mongolia

Sample Nos.	No. Group, Locality	*Age*, Lithology	Mean palaeomagnetic directions *		α_{95} (°)	k	n	Notes
			D (°)	I(°)				
		Cretaceous						
M 86–M 90	12 Chulut Tsagan Del	basalts	175.8	−62.8	5.6	41.01	17	Thermal cleaning
M 91–M 95	13 Chulut Tsagan Del	basalts	167.1	−63.7	6.1	31.03	19	Thermal cleaning
M 201–M 219	14 Gobi desert	sandstones	0.2	72.0	2.7	105.40	27	Thermal cleaning,
			202.7	−58.7	5.8	21.76	30	multi-component analysis
		Permian						
M 1–M 15	1 Barun Buren	andesites						
M 16–M 31	2 Barun Buren	andesites	349.2	56.5	3.7	28.24	55	Thermal cleaning,
		tuffs	83.7	−39.2	4.5	153.3	8	multi-component analysis
M 32–M 43	3 Barun Buren	tuffs						
M 44–M 49	4 Amarchid	tuffs						
M 50–M 56	5 Tel Tin Gol	tuffs	314.8	36.7	4.1	88.5	15	Thermal cleaning,
			157.3	−13.1	5.2	86.03	10	multi-component
M 96–M 100	6 Chulut Tsagan Del	tuffs	83.0	−34.1	11.3	9.82	19	analysis
M 101–M 105	7 Chulut Tsagan Del	tuffs						
M 109–M 113	8 Chulut Tsagan Del	andesites						
M 81–M 85	9 Chulut Tsagan Del	sandstones						
M 106–M 108	10 Chulut Tsagan Del	silicic andesites						
		Carboniferous						
M 225–M 233	16 Gobi–Mandach	tuffites	350.6	−10.6	5.1	23.59	35	A.F. cleaning
M 234–M 237	16 Gobi–Mandach	tuffites	328.4	9.5	3.6	121.2	14	A.F. cleaning
M 238–M 240	16 Gobi–Mandach	tuffites	338.9	−8.5	2.3	454.6	10	A.F. cleaning

* Corrected for bedding tilts.

clastic tuffs to fine-grained agglomerates. The rocks belong to the tholeiitic series with a distinct calc-alkaline tendency. They are frequently characterized by brownish and reddish, rarely by violet colours related to the higher percentages of Fe (Bajlich et al., 1986).

The Cretaceous volcanics, basalts and andesites form extensive lava flows overlying the conglomerate series of the Nilgin Cretaceous depression which contains pebbles of Jurassic Bajan–Cagan granite. Chemical analysis indicates that they correspond to andesite basalts or to even more acidic rocks. This is a continuous series that occurs in the whole of central Mongolia (Špaček and Abraham, 1981).

The Barun Buren area and the Tel Tin Gol locality are in the Selengin volcano-plutonic zone of northern Mongolia. The zone is a product of the Upper Palaeozoic and Mesozoic stage of orogenic activation of the Lower Caledonian bedrock, it is elongated in the W–E direction over a length of more than 1000 km and its width varies from 100 to 250 km. Its evolution occurred during three periods: the Permian, Triassic–Lower Jurassic and Jurassic. The emergence of a volcanogenic series in intimate comagmatic relationship with an extensive complex of granitoids of various basicities is common to all three periods. The Permian volcanism is responsible for the generation of major systems of lave flows of basalt andesites and liparities, tuffs, tuff breccias, with tuffogenic sedimentary rocks in places. At the beginning of the Permian, mostly dark grey, greenish grey and violet lavas, frequently with amygdaloidal andesites, andesite basalts and porphyric basalts, accumulated over most of the region. Acidic rocks occur at various stratigraphic levels, but mainly in the upper part of the system. The Middle Permian

is mostly characterized by a tuffogenic series composed of grey, grey-green tuffites, tuffo-aleurolites, tuffs and tuff conglomerates and rare lava intercalations of acid and intermediate composition.

After the accumulation of the tuff series, the volcanic activity east of the Orchon fault increased during the second half of the upper Permian. In the central part of the zone, between the rivers Orchon and Selenga, the volcanics are of mixed composition and vary from basalts and andesites to liparities (Čuta et al., 1981).

The oriented hand samples of Cretaceous basalts (M86–M95) (Table 1) are from the northern margin of the upper part of the Nilgin Cretaceous depression south of the Chulut Tsagan Del ore deposit (Špaček and Abraham, 1981). The geological age of these rocks is variable; for example, the outcrops found 200 km southeast of the sampling locality are claimed to be Upper Jurassic. The sampled rocks belong to the lower part of the volcanic phase of one flow, and have been determined petrographically as andesite basalts. They are porous amygdaloidal lavas belonging to shallow-water and terrestrial effusions. The dip of the rocks at both the localities does not exceed 10° S.

In the southern part of Mongolia, in the Gobi Desert, samples of Cretaceous sandstones (M201–M219) (Table 1) were collected from a cross-section 20 m thick and about 300 m long, whose dip did not exceed 15°.

The Permian volcanic rocks collected from the northern part of Mongolia are amygdaloidal andesites (M1–M15), andesite–pyroclastic tuffs (M16–M43) and silicified andesite tuffs (M44–M56) (Table 1).

The andesite tuffs and dacites of central Mongolia are of Middle–Upper Permian age. Their chronology was determined with reference to granite intrusions dated as Lower Jurassic by analogy to the age of the other granite massifs of Mongolia. Samples of andesite tuffs (M96–M105) and fluidal andesites (M109–M113) (Table 1) were collected from near the Chulut Tsagan Del deposit.

Carboniferous tuffites (M225–M240) were collected from three outcrops over an area of 200–250 m at the Mandach locality in the Gobi Desert, southern Mongolia.

Laboratory tests for palaeomagnetic stability

Pilot samples were selected from each locality and subjected to laboratory tests for stability using an alternating field up to an maximum amplitude of 140 mT and by heating in a magnetically compensated medium (Krs, 1969).

In the course of alternating field demagnetization the specimen was rotated at a ratio of angular velocities of 5 : 4, each specimen being placed in the demagnetization coils in two mutually perpendicular positions. In the thermal demagnetization experiments, the specimens were heated in a non-magnetic furnace to the Curie point of haematite. The compensation of the magnetic field in the Helmholtz coils was controlled by the MAVACS system (designed by J. Halousek, B. Pešina and K. Příhoda) which guarantees automatic compensation to within 2 nT. The studied rocks were measured using highly sensitive spinner magnetometers of the JR-4 type (Jelínek, 1966). Each heating cycle was followed by a determination of the mean magnetic susceptibility in order to record any mineral phase changes which may have occurred in the rock. This measurement was made with a KLY-2 bridge (Jelínek, 1973).

NOTATION

In this paper the SI system of units has been used, i.e. (nT) for the remanent magnetization (RM) and 10^{-6} (SI) for the mean magnetic susceptibility. The abbreviations and symbols used are below:

NRM natural remanent magnetization
M_0 modulus of the remanent magnetic moment in natural conditions.
M_t the modulus after the thermal field has been applied.
$M_{\bar{H}}$ the modulus after the alternating field has been applied.
H intensity of the A.C. magnetic field.
κ_0 the mean magnetic susceptibility measured at 20° C.
κ_t the mean magnetic susceptibility measured after the thermal field has been applied.

Stereographic projections have been used to plot the directions of the RM vectors. The black dots (triangles) represent projection onto the lower hemisphere while the open dots (triangles) represent projections onto the upper hemisphere. The sets of samples were treated by Fisher statistics (1953); α_{95} is the semi-angle of the cone of confidence at 95% probability, k is the precision parameter, n the number of speci-

mens analysed and N the number of localities involved in the palaeomagnetic interpretations.

In the stereographic projection the crossed black (open) dot (triangle) represents the mean direction of the set and the circle marks the trace of the cone of confidence. The asterisk indicates the direction of the field of the present co-axial geocentric magnetic dipole.

Cretaceous basalts and sandstones

The Cretaceous basalts from localities 12 and 13 in the neighbourhood of Chulut Tsagan Del were treated for stability using an alternating field. Minor changes in the RM directions were observed, an-hysteretic magnetization was displayed by some specimens at higher field intensities during demagnetizations, while in other specimens not even the maximum field applied was able to separate the stable component of the RM vector.

Pilot samples of Cretaceous basalts from the same localities were demagnetized thermally (Fig. 1). The graphs of the normalized moments vs. temperature indicate that magnetite, or titanomagnetite is the carrier of the magnetically stable RM. Both groups of Cretaceous basalts were cleaned thermally. Table 1 summarizes the statistical parameters of the sets of samples collected at both localities by the appropriate applied thermal fields. Characteristic thermal demagnetization graphs and component diagrams are displayed by sandstone samples M203/1 with a normal RM direction and also by M215/1 and M217/2 with reverse RM directions (Fig. 2). The whole set of samples was demagnetized by a thermal field. The analysis of M_t/M_0 curves, RM directions and multicomponent analysis (Kent et al., 1983) separated the sandstone samples from the same outcrop (including sample M217) into groups with normal and reverse RM directions (Table 1); only the reverse polarization is interpreted as primary magnetization.

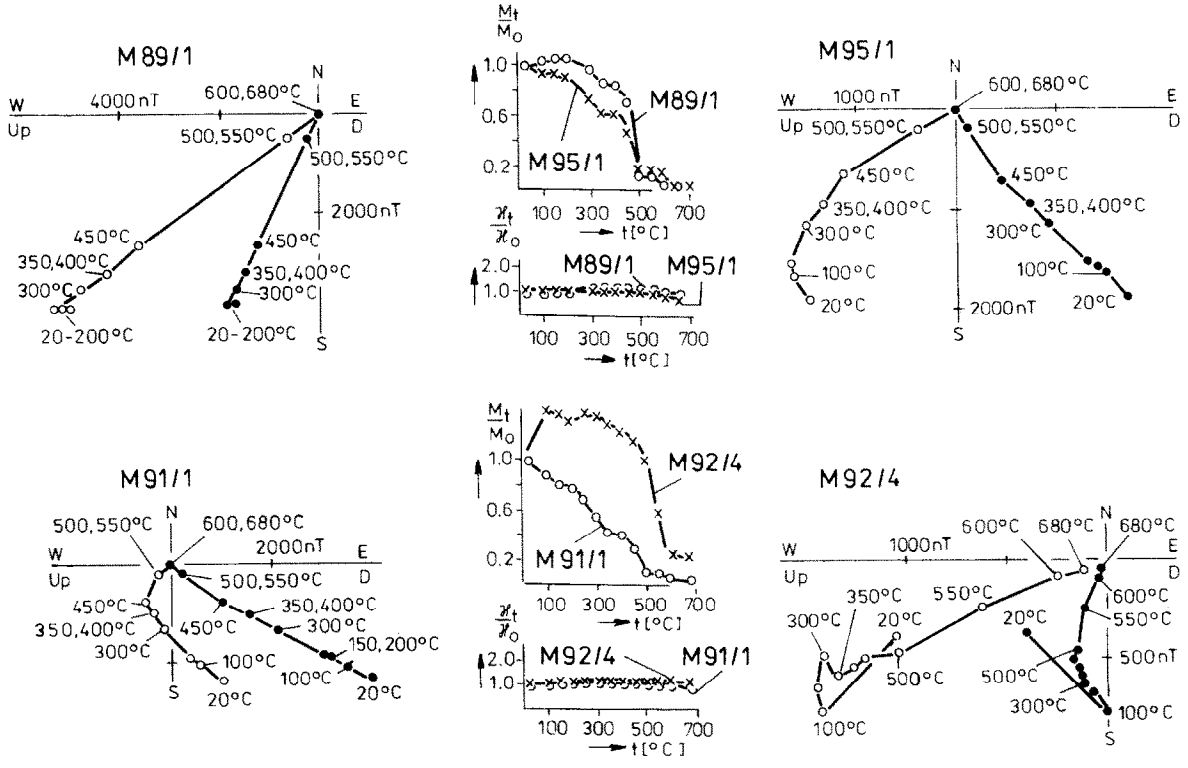


Fig. 1. Result of partial thermal demagnetization of Cretaceous basalts from the Chulut Tsagan Del locality, sites 12 and 13.

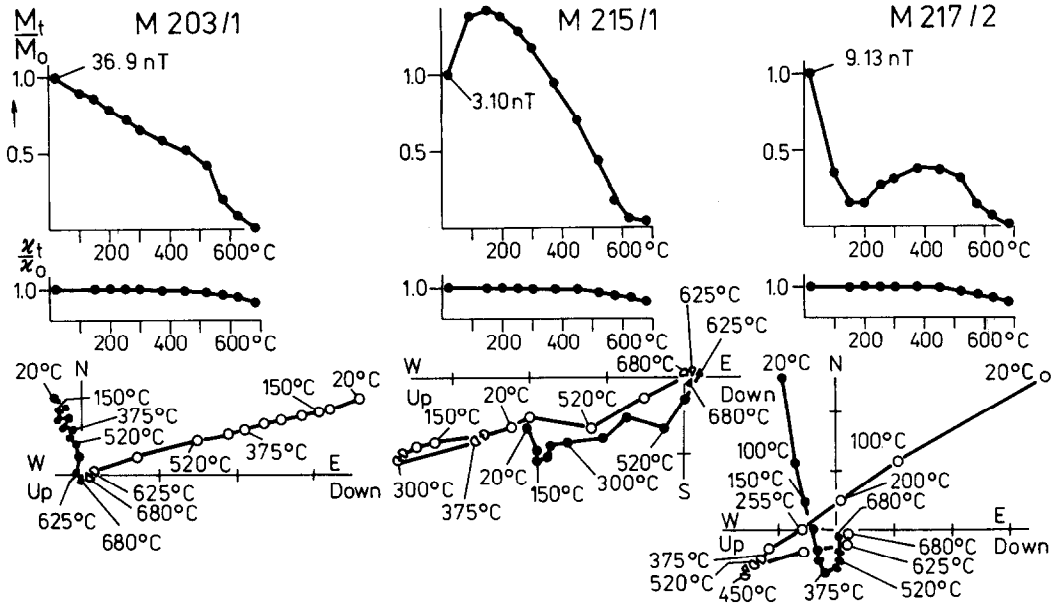


Fig. 2. Thermal demagnetization of Cretaceous sandstones from the southern part of Mongolia, Gobi Desert, site 14.

Permian andesites and tuffs

The pilot samples of Permian andesites and tuffs were A.F. and thermally demagnetized. The stability with respect to alternating fields was studied in seven samples and mostly minor changes in the RM directions were observed.

Thermal demagnetization of the andesites proved to be more effective. The samples from groups 1, 3, 4, 7, 8, 9 and 10 (Table 1), which were

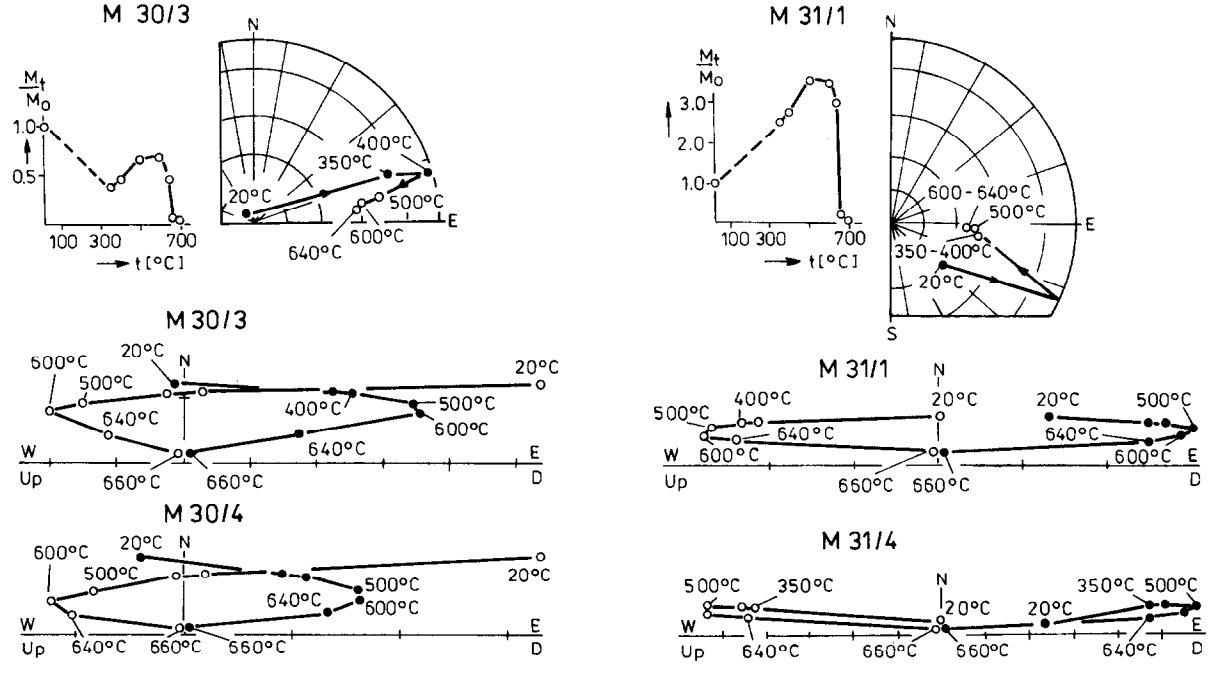


Fig. 3. Thermal demagnetization of Permian andesite tuffs from the Barun Buren—reverse group, site 2.

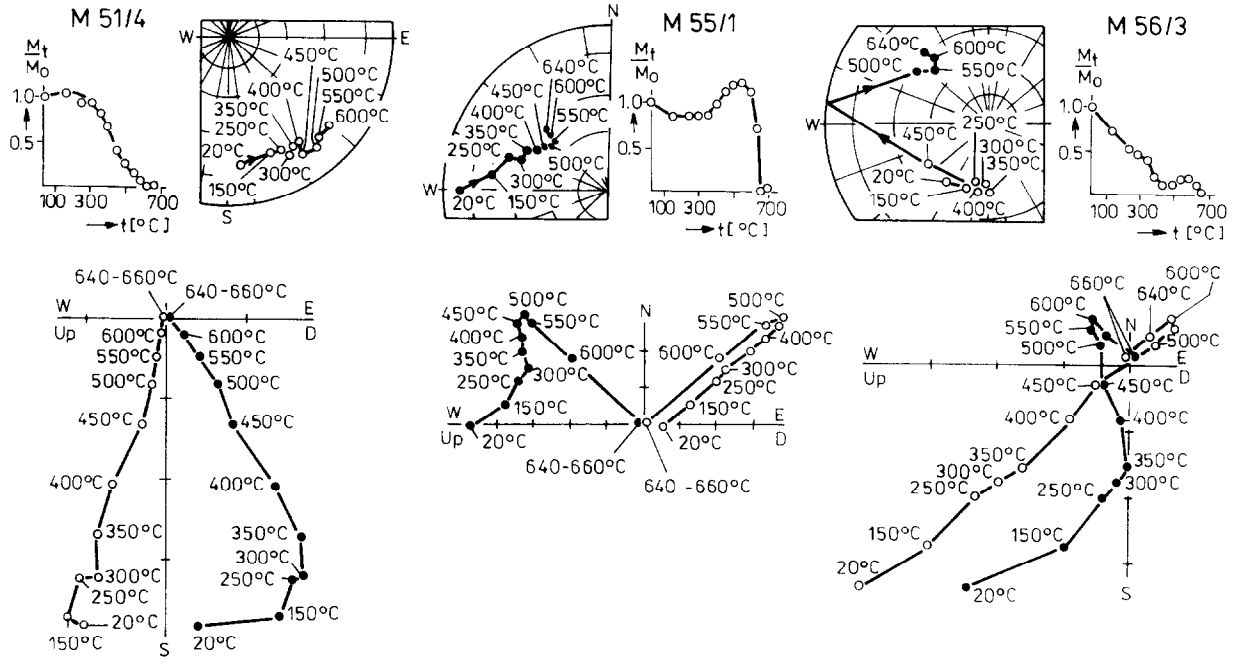


Fig. 4. Permian andesites, three different types of thermal demagnetization graphs, RM directions and component diagrams from the Tel Tin Gol locality.

magnetically soft and strongly unstable or remagnetized, proved to be unsuitable. The high magnetic purity of the demagnetizer (better than 2 nT) enabled a number of components of magnetization to be determined in a single rock sample.

In the course of thermal cleaning of the samples of andesite tuffs M16–M31 from the Barun Buren locality in northern Mongolia, groups with normal and reverse RM directions were determined (Fig. 3). Both groups display one carrier of magnetization and a Curie temperature corresponding to haematite. The multicomponent and RM direction analyses indicate that the group with normal RM directions ($D = 349°$; $I = 56°$) was probably remagnetized during the Jurassic or Cretaceous. The statistical parameters of both groups are given in Table 1.

The silicified andesite tuffs of the Early Permian, samples M50–M56 from the Tel Tin Gol locality, were also demagnetized thermally (Fig. 4). Three different types of thermal demagnetization curves, RM directions and component diagrams (Fig. 4) were isolated from this group. Sample M51/4 displays reverse RM directions and a Curie temperature corresponding to magnetite (or titanomagnetite). Sample M55/1, on the other hand, displays normal RM directions and a Curie point corresponding to haematite. The thermal demagnetization curve of sample M56/3 indicates that magnetite and haematite are present. With increasing temperature, the reverse RM direction changes to normal, and the component diagram displays three magnetization components. The carrier of palaeomagnetization in the rock is presumed to be reversely polarized magnetite and normally polarized haematite.

The microscopic study was carried out by Dr. M. Klečka of the Institute of Geology and Geotechnics of the Czechosl. Acad. Sci.: Sample M51—magnetite (possibly Ti-magnetite) is mostly made of hypidiomorphic grains, part of which are affected by magnetic corrosion. The corrosion is uniformly distributed throughout the rock; in detail it is bound to the fringes of plagioclase phenocrysts and dark minerals.

Haematite is represented in substantially smaller amounts in the form of xenomorphic irregular clusters close to magnetite phenocrysts. It does not replace magnetite immediately, but rather forms an independent (weakly anisotropic) phase.

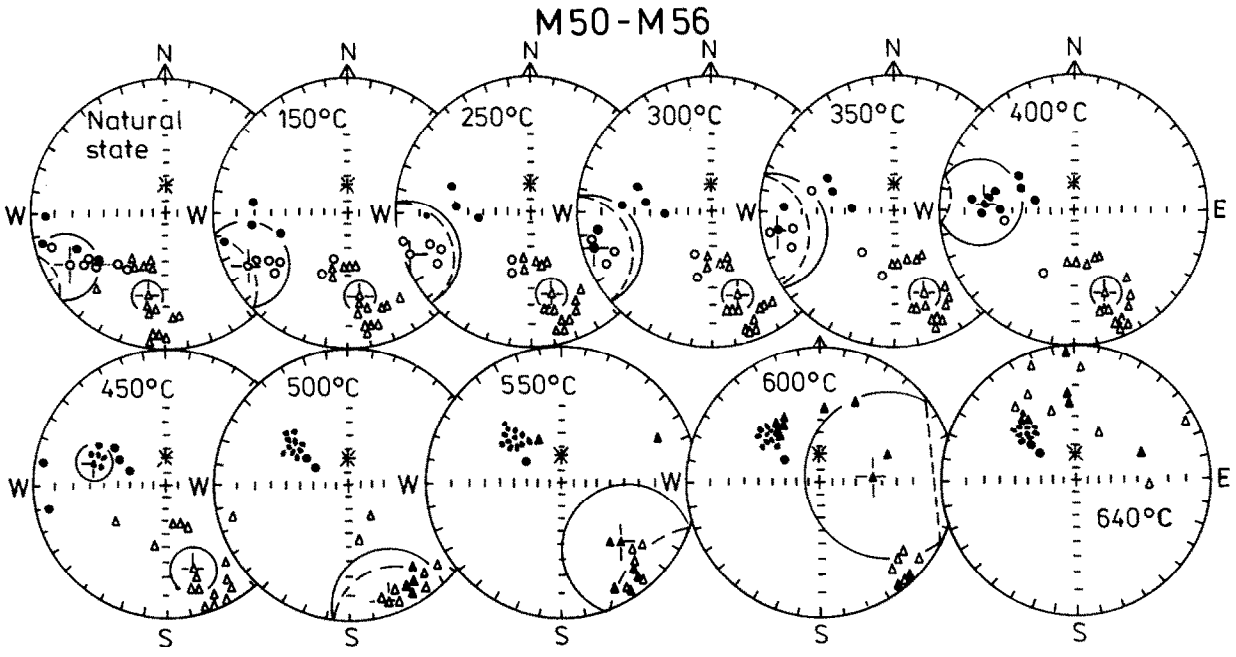


Fig. 5. The directions of remanent magnetization of thermally demagnetized samples of Permian andesite tuffs from the Tel Tin Gol locality, site 5.

In sample M55 haematite is weakly anisotropic and occurs in the form of hypidiomorphic grains; this apparently involved the pseudomorphosis of haematite replacing magnetite. Certain domains within the grains are still isotropic, i.e. negligible relicts of magnetite. The grains are square to hexagonal in cross-section.

In sample M56 magnetite forms hypidiomor-

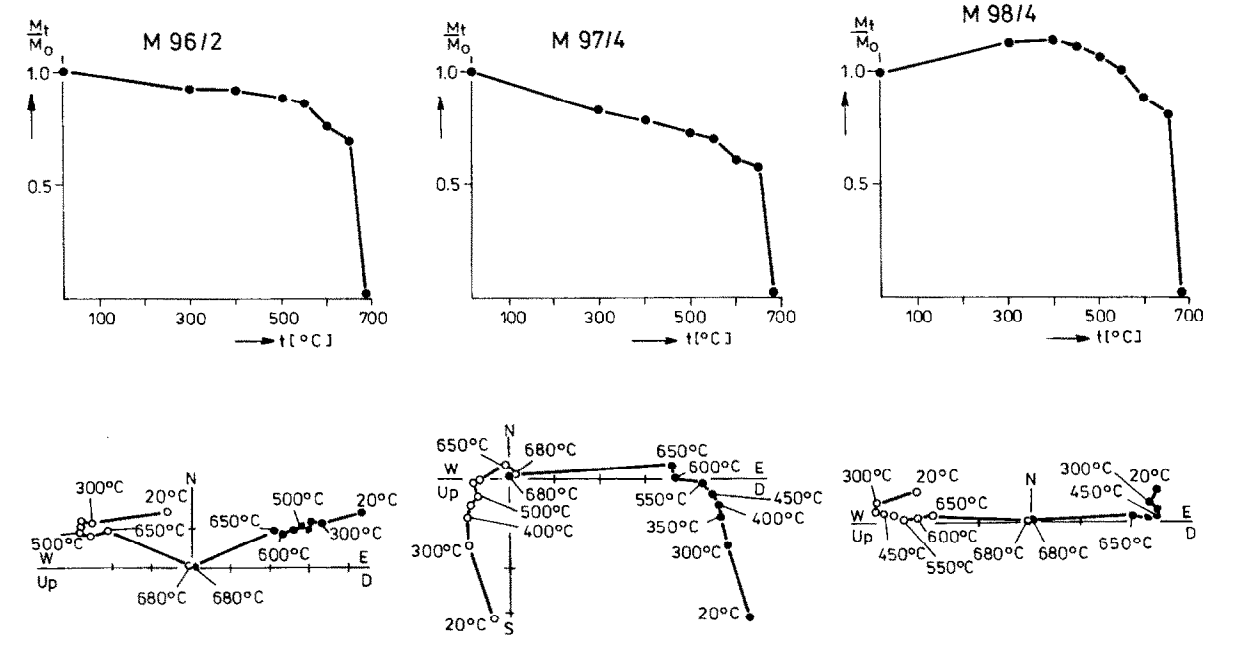


Fig. 6. Thermal demagnetization of Permian andesites from the Chulut Tsagan Del locality, site 6.

TABLE 2

Palaeomagnetic directions and pole positions for Cretaceous, Permian and Carboniferous rocks in Mongolia

Sample Nos.	Age *	Geographic coordinates		Mean palaeomagn. directions		α_{95} (°)	k	n	N	Percentage of polarity		Palaeomagnetic pole positions		Ovals of confidence	
		lat. (°N)	Long. (°E)	D (°)	I (°)					R	N	lat.	long.	dm (°)	dp (°)
M 86–M 95															
M 201–M 219	K	45.386	107.551	187.7	−61.1	4.2	18.59	66	3	100	0	82.91°S	41.70°E	6.4	4.9
M 16–M 31															
M 96–M 100	P_3	48.123	105.958	83.2	−37.8	7.8°	13.80	27	2	100	0	11.29°N	35.19°E	9.1	5.4
M 50–M 56	P_1	49.500	105.133	324.8	27.8	6.4°	21.28	25		60	40	45.01°N	22.92°W	7.0	3.8
M 16–M 31															
M 50–M 56															
M 96–M 100	P	48.280	106.000	295.1	37.0	8.5°	6.43	52	3	71	29	31.83°E	11.83°N	9.9	5.8
M 226–M 240	C	44.649	108.250	345.6	−5.4	3.9°	23.25	59	3	100	0	40.92°N	52.60°W	3.9	2.0

* K = Cretaceous, P_3 = Late Permian, P_1 = Early Permian, P = Permian, C = Carboniferous.

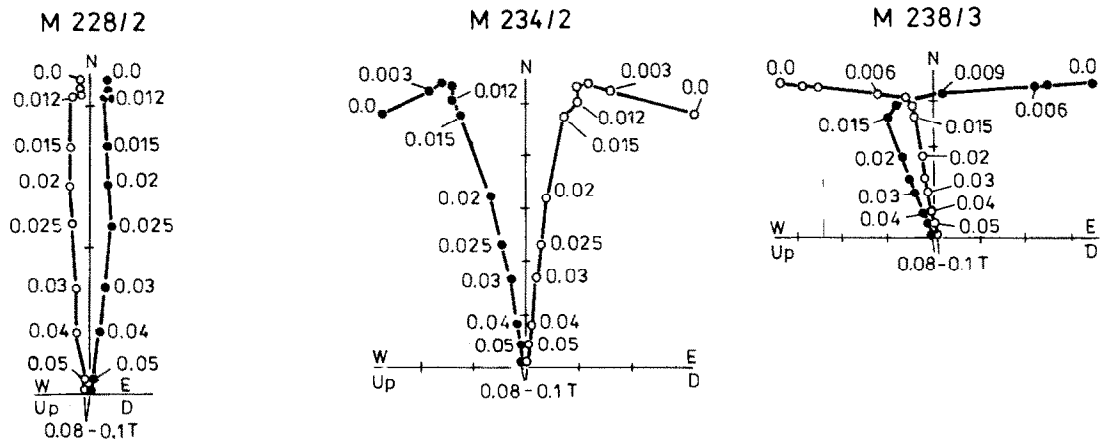


Fig. 7. Result of alternating field demagnetization of Carboniferous tuffites from the Gobi Desert, Mandach locality, site 16.

phic grains, otherwise the same applies as for sample 51. Haematite forms irregular xenomorphic grains in the immediate vicinity of magnetite phenocrysts. On the one hand, it forms an independent phase, on the other, in several cases haematite formation occurs along cracks of crushed magnetite phenocrysts (magnetite–haematite alteration). It appears that haematite crystallized as one of the last minerals, or it could also have originated in the later stages of the

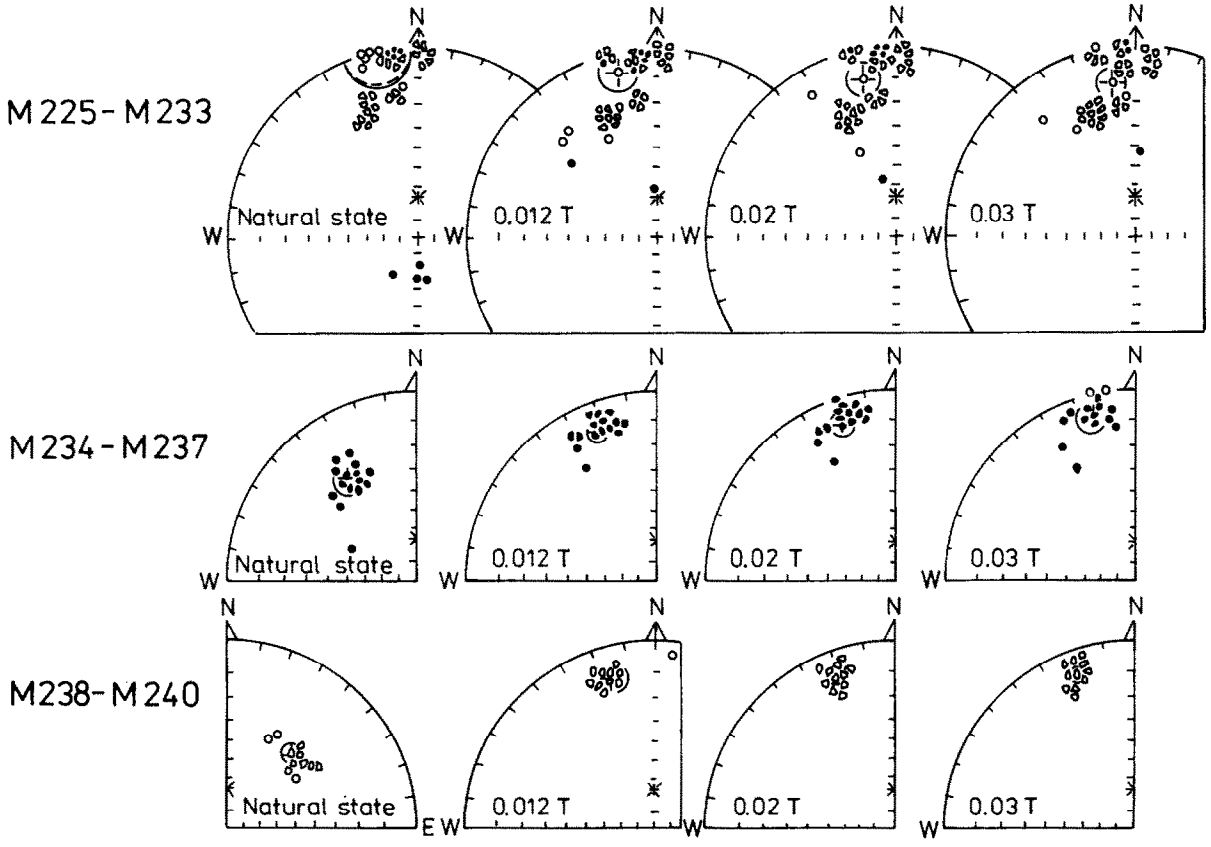


Fig. 8. Alternating field demagnetization of Carboniferous tuffites from three outcrops in the Gobi Desert, Mandach locality, site 16.

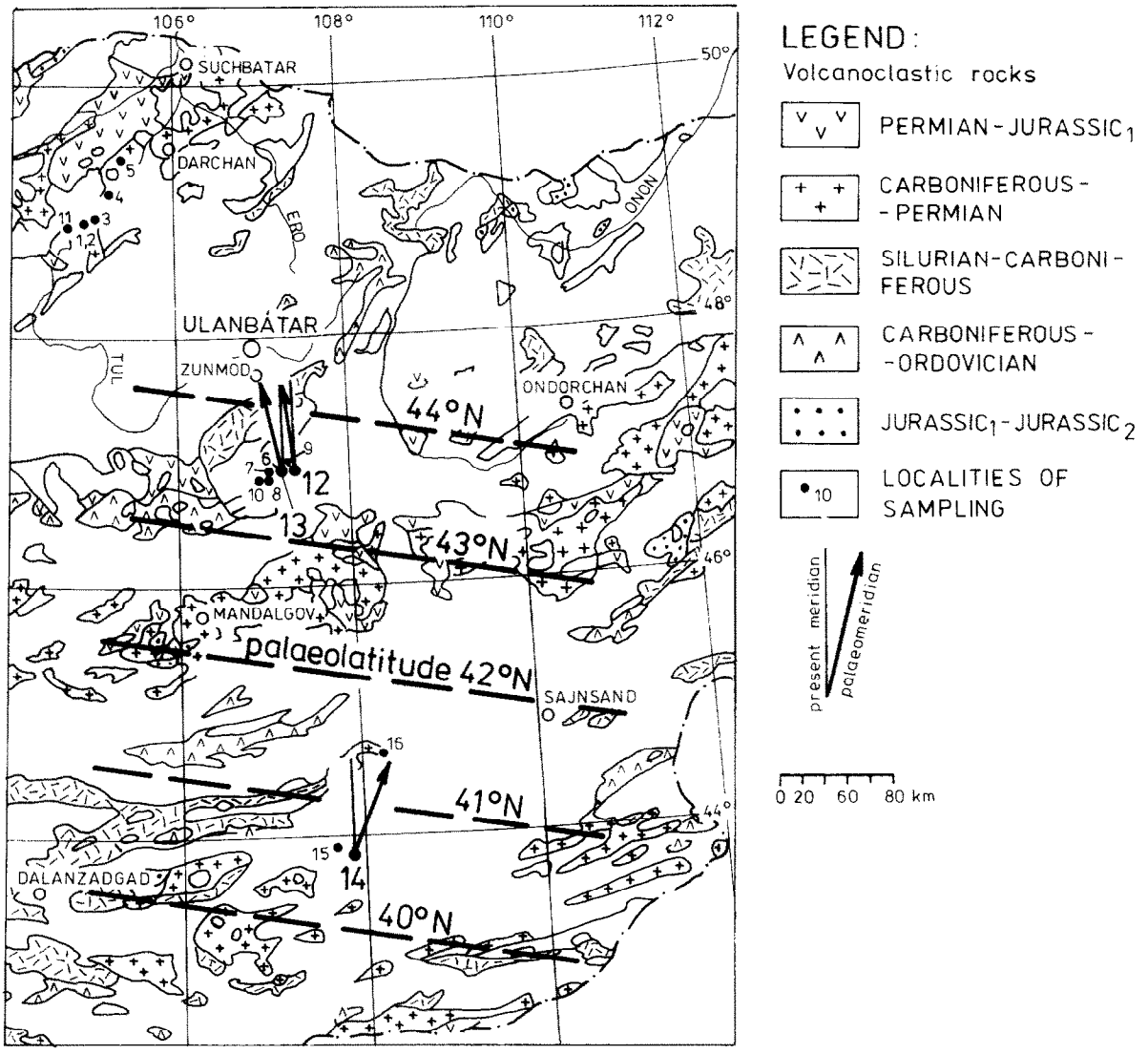


Fig. 9. Cretaceous palaeomagnetic declinations and palaeolatitudes for the central part of Mongolia.

evolution of the rock, perhaps in the hydrothermal stage of the evolution of volcanics.

The above information implies that an Early Permian group with reverse directions of RM and a group with normal RM directions, which dates to the Early to Middle Permian were found in the same Tel Tin Gol outcrop (M50–M56, Figs. 4, 5 and Table 2).

The RM directions of the demagnetized samples of andesite tuffs M96–M100 from the Chulut Tsagan Del locality in central Mongolia, which were thermally treated, Fig. 6, correlate well with the reverse group M16–M31 (Barun Buren in northern Mongolia). The resultant palaeomagnetic parameters are given in Table 1.

Carboniferous tuffites

Pilot samples of the sample set M225–M240 from three outcrops in the southern part of Mongolia (Mandach in the Gobi Desert) were subjected to laboratory stability tests using alternating and thermal fields. The results indicate that both methods can be used, but the more effective as regards cleaning is the A.F. demagnetization method (Figs. 7 and 8).

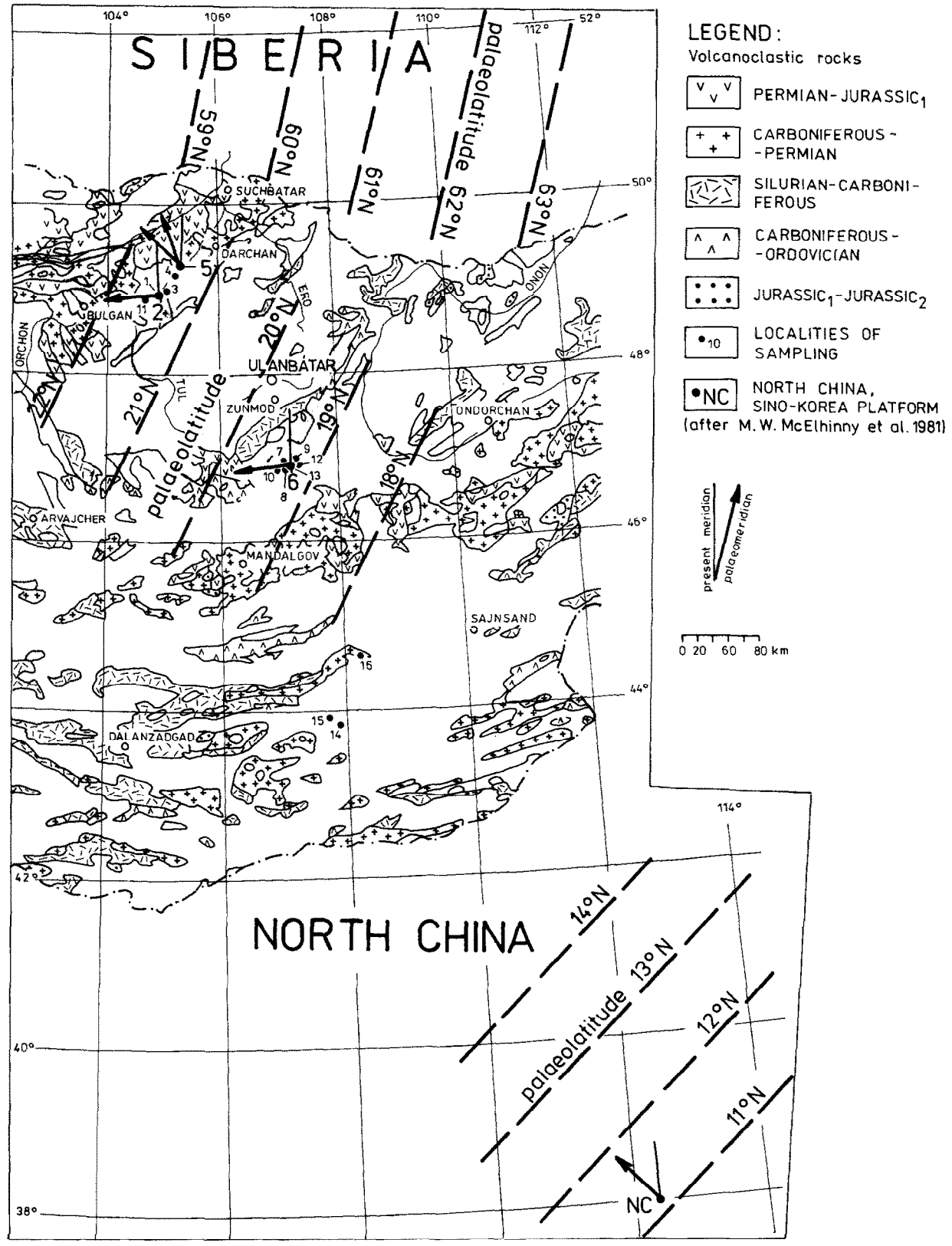


Fig. 10. Permian palaeomagnetic declinations and palaeolatitudes for the central part of Mongolia and Permian palaeolatitudes for central Siberia and the Sino-Korean block (northern China).

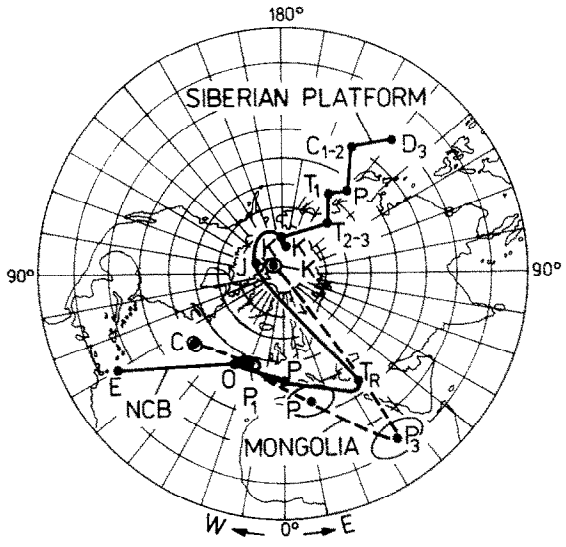


Fig. 11. Palaeomagnetic pole for the Cretaceous and Carboniferous, and virtual pole positions for the Permian (P_1–P_3) of Mongolia. APWPs for North China (NCB) (Lin et al., 1985) and Siberia platform (Khramov, 1982).

Table 1 gives a summary of the statistical parameters calculated for the alternating fields used and the separate groups of samples of Carboniferous tuffites.

The main results

The preliminary results of the palaeomagnetic investigations of Cretaceous, Permian and Carboniferous rocks of Mongolia are presented in Table 2 and can be interpreted for this region relative to the Siberian platform and the Sino-Korean block (northern China).

(1) The results related to the Cretaceous basalts and sandstones of northern and southern Mongolia are interpreted as follows: in the Cretaceous this block was part of the Siberian platform and northern China. The coordinates of the palaeomagnetic pole are 83° N and 222° E (Table 2) and the palaeolatitudes calculated for this region are shown in Fig. 9.

(2) The results of the palaeomagnetic investigations of Permian (P_1, P_3) andesites and tuffs from northern and central Mongolia (Table 2) indicate considerable differences in the palaeodeclination: 264° for P_3, 325° for P_1. The palaeomagnetic pole, calculated for the Permian, has the coordinates 32° N and 12° E. McElhinny et al. (1981) give a pole position for the Sino-Korean block at 44° N and 358° E. Khramov (1982) reports a pole for the East Siberian plate at 45° N and 141° E during the Permian. These results imply that northern and central Mongolia was probably a part of the Sino-Korean block (northern China) in the Permian. The present author calculated the palaeolatitudes for the region of central Siberia, including the Altai–Saian block, bordering Mongolia in the north, using the data of Khramov (1979) and Irving et al. (1976), and for the Sino-Korean block from the values given by McElhinny et al. (1981). The palaeolatitudes calculated for the Permian and shown in Fig. 10 distinctly indicate that the Siberian plate differed in palaeolatitude and palaeomeridian from northern and central Mongolia. Good correlation, however, can be observed between the data related to the Sino-Korean block and the palaeolatitudes of Mongolia.

(3) The results of the Carboniferous tuffites yield a pole at 41° N and 53° W (Table 2), again this differs from the pole given for the Carboniferous of the Siberian plate, 30° N and 150° E (Khramov, 1982).

(4) To conclude, the different palaeogeographic evolutions of the studied region of Mongolia and the Siberian plate during the Carboniferous and Permian, can be seen from the apparent polar wander path (Fig. 11). The polar wander paths for northern China (Lin et al., 1985) and northern and southern Mongolia, however, are similar.

These preliminary results indicate that more suitable rock types will have to be sampled for the Jurassic to Carboniferous with the aim of understanding the formation of the Central Asian region in time and space.

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